Type MS · Explain eye defects and how they are corrected or treated

Answer 1 of 11 · Question 8 part b.iii · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Question number					
8	(a)	(i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
		(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
	(b)	(i)	behind the retina/fovea;		1 mark
		(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
		(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
	(c)		• infection; • rejection;		2 marks

Total 13 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 1 of 11 · Question 8 part b.iii · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

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Type MS · Explain eye defects and how they are corrected or treated

Answer 2 of 11 · Question 3 part c · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

3	(a)	(i)	A (cornea and lens); B is incorrect as the iris does not play a role in focusing light C is incorrect as the iris does not play a role in focusing light D is incorrect as the pupil does not play a role in focusing light		1
	(b)	(i)	Points as follows: 150:2.7 200:3.2 250:2.3;		1
Clip (b)(i) with (b)(ii) and (b)(iii)		(ii)	Line of best fit missing out point at 150:2.7	Ignore lines that extended beyond points	1
		(iii)	150 lux/150:2.7/2.7 diameter;		1
			does not follow the pattern of other points/trend/not on the line of best fit;		1
		(iv)	As light intensity increases diameter decreases;	ORA	1
	(c)		Idea that distant/far away objects are blurred/difficult to see/ORA; as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much;	Reject can't see	1 1

Total question 3 = 8 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 2 of 11 · Question 3 part c · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 3 of 11 · Question 1 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

1(a)	line labelled A pointing to retina; line labelled B pointing to ciliary muscles; line labelled C pointing to iris; line labelled D pointing to optic nerve; line labelled E pointing to eyelid;		5

Question Number	Answer	Additional guidance	Mark
1(b)	bending at cornea; bending at lens; image formed on retina;		**3**

Question Number	Answer	Additional guidance	Mark
1(c)	long sight/myopia; short sight/hypermetropia; astigmatism;		3

Total 11 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 3 of 11 · Question 1 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain eye defects and how they are corrected or treated

Answer 4 of 11 · Question 8 part b.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
8	(a)	(i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
		(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
	(b)	(i)	behind the retina/fovea;		1 mark
		(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
		(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
	(c)		• infection; • rejection;		2 marks

Total 13 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 4 of 11 · Question 8 part b.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Explain eye defects and how they are corrected or treated

Answer 5 of 11 · Question 8 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
8	(a)	(i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
		(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
	(b)	(i)	behind the retina/fovea;		1 mark
		(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
		(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
	(c)		• infection; • rejection;		2 marks

Total 13 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 5 of 11 · Question 8 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Explain eye defects and how they are corrected or treated

Answer 6 of 11 · Question 6 part c · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

6	(a)	(i)	• A=rods/cones/retina; • detects light; • converts light energy into electrical energy; • (changed into (nerve) impulses; • B = transmits (nerve) impulses/electrical signals; • from (retina) to brain;	At least 1 mark from A and B to get full marks	4
		(ii)	• bent inwards at cornea/lens; • rays pass through the cornea and converge after the lens; • rays cross in eye (after lens); • inverted image/arrow;	Do not accept bending outwards for MP1 and MP2 Mp3- do not accept rays crossing before the retina	4
	(b)		• use both eyes and one eye; • use of eye patch/eq; • ball thrown same distance/stated distance;; • ball thrown many times; • count how many times ball is caught; • same type/size of ball;		4
	(c)		• misshapen/eq. cornea/lens; • light not focussed correctly/ light scatters/light not refracted properly; • blurred image;		2
					Total =14

Type MS · Explain eye defects and how they are corrected or treated

Answer 6 of 11 · Question 6 part c · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 7 of 11 · Question 8 part c · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number					
8	(a)	(i)	• arrow heads showing direction of light rays; • bent at cornea; • bent at lens; • cross over in eyeball; • inverted image on retina;		5 marks
		(ii)	transmits/carries impulses from the retina/rod and cones; to the brain;		2 marks
	(b)	(i)	behind the retina/fovea;		1 mark
		(ii)	eyeball too short / eye lens not powerful enough/too flat/not convex enough;		1 mark
		(iii)	use a convex / converging lens; to converge/refract/bend rays of light;		2 mark
	(c)		• infection; • rejection;		2 marks

Total 13 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 7 of 11 · Question 8 part c · 2 marks

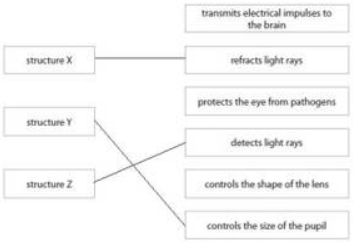
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Type MS · Explain eye defects and how they are corrected or treated

Answer 8 of 11 · Question 1 part b.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

1	(a)	(i)			3
		(ii)	A; B, C and D do not describe stereoscopic vision.		1
	(b)	(i)	build-up of protein/cloudy/opaque; in the lens;		2
		(ii)	surgery/operation to replace the lens/lens replacement;		1
		(iii)	39 000000 x 43 ÷ 100; 16.77 million/16.8 million/16 770 000;		2
		(iv)	vitamin A;		1

Total 10 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 8 of 11 · Question 1 part b.i · 2 marks

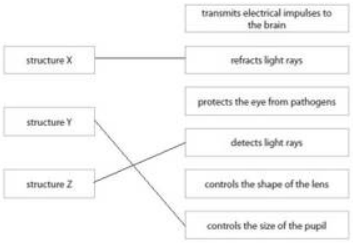
2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 9 of 11 · Question 1 part b.ii · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

1	(a)	(i)			3
		(ii)	A; B, C and D do not describe stereoscopic vision.		1
	(b)	(i)	build-up of protein/cloudy/opaque; in the lens;		2
		(ii)	surgery/operation to replace the lens/lens replacement;		1
		(iii)	39 000^{0}000 x 43 ÷ 100; 16.77 million/16.8 million/16 770 000;		2
		(iv)	vitamin A;		1

Total 10 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 9 of 11 · Question 1 part b.ii · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 10 of 11 · Question 3 part c · 1 mark

2024 JUN · Paper 02 · find part c in this mark scheme extract

Number			
3 (a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 11 of 11 · Question 3 part a.iv · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

3 (a)	(i)	Part W - pupil; Part X – retina;	1 mark for each correctly named part	2
	(ii)	• sends (electrical) impulses from photoreceptors/rods/cones; • to the brain/CNS; OR • sends (electrical) impulses from brain/CNS; • to iris/ciliary muscles;		2
	(iii)	lens becomes thinner/flatter/less convex; less refraction;	allow less light refracted	2
	(iv)	Any two from: • shape of cornea restored; • light refracted, onto the retina/correctly/notmally; • correct information sent to the brain; • image/vision not blurred/clear;		2
(b)		• measure the size/diameter of the pupil; • in a dark room; • shine a light into the eye; • measure/compare size/diameter of the pupil;	accept ref to different light intensities	4

Total Marks 12

Type MS · Explain eye defects and how they are corrected or treated

Answer 11 of 11 · Question 3 part a.iv · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks